Maths Exploring the Terminal Assessment-07

QCM A · conditioning and independence

After S37.

For each question, only one proposal is accurate.
consult the correction.

1. $P_A(B)$ means:

A. $P(A \cap B)$ B. $P(B \mid A)$

C. $P(A \mid B)$ D. $P(\overline{B})$

2. If $P(A) = 0.4$ and $P_A(B) = 0.5$, then $P(A \cap B)$ is valid for:

A. 0,9 B. 0,2

C. 0,5 D. 0,1

3. Independent events verify:

A. $P(A \cap B) = 0$ B. $P(A \cap B) = P(A) + P(B)$

C. $P(A \cap B) = P(A)P(B)$ D. $A = B$

4. "Up to 3 hits" means:

A. $X < 3$ B. $X \leq 3$

C. $X \geq 3$ D. $X > 3$

5. In a weighted tree, the probability of a path is:

A. the sum of its branches B. the product of its branches

C. Always $\frac{1}{2}$ D. a difference

QCM B · binomial distribution and threshold

After S41 ; whole counting variables.

For each question, only one proposal is accurate.
consult the correction.

6. $X \sim \mathcal{B}(10;0{,}3)$ account:

A. the success of ten identical and independent tests B. Ten probabilities

C. Failures only D. a continuous average

7. For $X \sim \mathcal{B}(n;p)$, $E(X)$ is valid for:

A. p^n B. np

C. $n + p$ D. $n(1-p)$

8. "Strictly more than 4" means:

A. $X \geq 4$ B. $X \leq 4$

C. $X \geq 5$ D. $X > 5$

9. Not all the assumptions in Bernoulli's scheme are satisfied if:

A. there is a fixed number of tests B. Success has the same probability

C. The tests are dependent D. Outcome is success/failure

10. To check a minimum integer threshold in a monotonous problem, we also check:

A. Previous ranking B. A tree colour

C. $P(X = 0)$ only D. the name of X

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $P_A(B) = P_B(A)$ in any situation.

2. If $X \sim \mathcal{B}(n;p)$, then $P(X \geq k) = 1 - P(X \leq k-1)$.

3. An observed frequency is exactly equal to the theoretical probability.

Review review

I know how to build a tree, choose or refuse the binomial model and translate an integer threshold. **To be monitored:** Invert conditioning and intersection; forget the shift of an entire bound.